

Online Appendix to “Informal Regulation and Housing Supply: Evidence from Chicago’s Aldermanic Privilege”

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A Data and Sample Construction

This appendix provides additional detail on the data sources and sample rules described in Section 2.

A.1 Data Sources

The property data come from Cook County Assessor sales, assessment, parcel, and improvement files. RentHub provides rental listings from 2014–2022. Building permits and ward boundaries come from the City of Chicago. The amenity controls use school locations, Chicago Park District boundaries, major streets, CTA stops, and Lake Michigan geometry; CTA distances use the permanent station network in place during the listing month or on the sale date.

Demographic controls come from American Community Survey block groups and 2010 and 2020 census blocks. Because block groups do not follow ward boundaries, I assign their population and household counts to wards in proportion to the area that falls inside each ward. I use the resulting counts to calculate ward-level tenure, race, ethnicity, and education shares. The ward income measure is the household-weighted average of block-group median household income. Small overlaps in historical ward maps are divided proportionally so that no part of a block group is counted twice.

A.2 High-Discretion Permits

The permit records cover Department of Buildings permits since 2006. The high-discretion group contains the permit types defined in Section 2.2; low-discretion placebo permits are the remaining routine types, excluding signs. The event-study outcomes omit records marked cancelled, revoked, or suspended.

Processing time is the number of days between application and issuance. The processing-time regression excludes missing dates, negative durations, and same-day permits; same-day permits remain in the lagged ward-month permit count. I assign each permit to the ward containing its coordinates under the boundaries in force on its application date.

A.3 Permit Event-Study Panel

The treatment panel compares each census block’s ward before and after the 2015 redistricting. I assign a block to the ward covering the largest share of its area. Let $a_o(b)$ denote the alderman representing block b under the old map and $a_d(b)$ the alderman under the new map. Blocks moved to a different ward are classified using the difference between these aldermen’s stringency scores:

$$\Delta S_b = S_{a_d(b)} - S_{a_o(b)}.$$

Positive values indicate assignment toward greater stringency, negative values indicate assignment toward greater leniency, and unchanged blocks form the comparison group.

For this design, stringency scores use permits filed from 2006 through 2014. Both origin and destination scores are assigned using the incumbents serving in 2014. The main sample retains blocks only when both incumbents remained in office when the new map took effect, so treatment matches the change the block experienced without using the outcome of the 2015 election. Section C.4 adds wards with electoral turnover and interprets the resulting estimate as assigned rather than necessarily realized stringency.

I count permit applications by block and year from 2010 through 2020. The sample

includes blocks whose geographic center lies within 500 ft of a pre-redistricting ward boundary. For reassigned blocks, distance is measured to the boundary separating the old and new wards; for unchanged blocks, it is measured to the nearest boundary of the original ward. The estimation sample requires at least one high-discretion permit during event times $t = -5$ through $t = -1$.

A.4 New Construction

The construction sample combines Cook County residential improvement records with the separate Assessor records used for larger multifamily properties. I link records across parcel divisions, replacement PINs, and the two Assessor sources so that each project enters once. I then compare the resulting sample with every issued residential new-construction permit, adding completed projects absent from the Assessor files when units, building area, and lot area can be measured consistently.

Projects must have positive lot area, positive building area, at least one dwelling unit, and a construction year from 2006 through 2022. Assessor histories sometimes report different years or repeat whole-building characteristics across assessment cards. I use permit and completion dates to select the year that best matches completion, combine cards that clearly describe one building, and remove later PINs representing the same project. Where construction years or lot and building areas appear inconsistent in the Assessor files, I supplement these records with manual research using permits, property records, and other documented sources. The replication files record these decisions and their supporting evidence.

Locations first come from the Historical Parcel Universe in the construction year. When the original PIN is unavailable there, I use the current location of that PIN or a verified historical address. Each project retains one location, preventing predecessor and successor PINs from entering separately. Multifamily status primarily follows Assessor property class. When the reported unit count does not plausibly describe the whole building, permit descriptions and other property records determine the whole-building count, with these decisions

also documented in the replication files.

Each project receives the broad zoning group in force on June 15 of its construction year. I combine official City zoning maps from 2012, 2014, and 2016 with passed map amendments dating to 2006, use annual parcel maps to follow sites whose PINs change, and retain planned developments as a separate zoning group. The reconstruction covers every project in the main and placebo boundary samples and reproduces the classifications observed in later official zoning maps. Floor-area ratio (FAR) is total building area divided by lot area; dwelling units per acre (DUPAC) is the number of units divided by lot acreage. Projects are assigned to the ward map and alderman in office on June 15 of the reported construction year because the Assessor does not provide an exact completion date. I calculate ward assignments and distances in the Illinois State Plane coordinate system used for Chicago (EPSG:3435) and report distances in feet.

A.5 New-Construction Summary Statistics

Table 1 describes the citywide sample of 13,635 projects with both FAR and DUPAC available, including 2,582 multifamily projects.

A.6 Home Sales

The raw file contains 902,830 parcel-sale records in Chicago’s eight townships from 2006–2022. The transaction sample includes Assessor classes 202–210, 234, 278, and 295, which cover detached houses and townhouses, and class 211, which covers apartment buildings with two to six units. I exclude mixed-use class 212, condominiums, cooperatives, larger apartment buildings, residential land, and all other property classes.

Transactions must have a nominal price above \$10,000, not be explicitly identified as land-only, involve one parcel, use a warranty or trustee deed, and pass all three Assessor sale-quality flags: no duplicate sale of the same property within 365 days, no low-price flag, and no excluded-deed flag. When buyer and seller names are available, I exclude transfers

with matching names. These transaction rules leave 238,705 sales with valid months and full 14-digit property identification numbers (PINs).

Some older records without an IDOR-refined date report the first day of the month even though the exact day is unavailable. I exclude these sales when either adjoining ward changes aldermen during that month, or when the ward map changes during the month, because the record does not identify the relevant alderman or ward map. This removes 1,289 transactions citywide and leaves 237,416. I retain records with precise IDOR dates and unrefined dates that are not recorded on the first of the month.

I measure property characteristics using the Assessor record for the same property and tax year as the sale. I keep properties with one recorded building, positive building area, and an Assessor proration rate of one, which indicates that the record describes the entire property rather than a fractional share. These restrictions leave 233,744 sales. I then exclude 45 sales whose property records report more bedrooms than total rooms, leaving 233,699 sales. Missing apartment counts do not disqualify class-211 buildings because unit counts are not required by the price specification. When the sale and improvement files report different residential subclasses, I use the class recorded in the sale file for the property-class fixed effect. I express prices in 2022 dollars using the monthly Chicago CPI-U, with the mean 2022 index as the base, and do not trim or cap the price distribution.

I locate each transaction by matching its PIN to the parcel map for the year of sale; all 238,705 sales meeting the transaction-quality criteria receive an exact historical location. For retained transactions, ward, neighboring ward, and alderman assignments use the recorded sale date: transactions before May 18, 2015, use the 2003 ward map, and those on or after that date use the 2015 map. I calculate distance to the nearest boundary of the property's ward in the Illinois State Plane coordinate system. Distance is positive on the side represented by the more-stringent alderman and negative on the other side. I drop boundaries if either alderman lacks a stringency score or the two scores are identical.

For the boundary-price design, I split ward-pair boundaries into approximately 1,320-ft

segments and assign each sale within 1,500 ft to the nearest segment on its ward-pair boundary. To compare similar homes in similar locations, the regression controls for log building area, log land area, log building age, log bedrooms, log bathrooms, a garage indicator, and distances to the nearest school, park, major road, CTA station open on the sale date, and Lake Michigan. Building age is sale year minus reported year built, and total bathrooms equal full bathrooms plus one-half of half bathrooms. The logged property characteristics must be positive, so properties sold in their reported construction year are excluded because their measured age is zero. The preferred 500-ft sample contains 58,468 sales. Property-class fixed effects compare sales within the same Assessor class, while segment-by-sale-quarter fixed effects compare sales near the same short stretch of boundary in the same three-month period. Standard errors are clustered by ward pair.

A.7 RentHub Listings

The rental data record each occasion on which RentHub observed a listing. I standardize addresses and building types and treat blank or placeholder addresses as missing. Because RentHub does not report a consistent unit identifier, I group observations into properties using the standardized address and rounded coordinates, or coordinates alone when the address is missing. Within each property, I identify a floorplan by its bedrooms, bathrooms, square footage, and building type. I take the median asking rent when the same floorplan appears more than once on a date and then take the median across dates within each month. Before monthly aggregation, rent and square footage are trimmed at the 1st and 99th percentiles within year-by-bedroom cells.

The sample requires positive rent, valid coordinates, and assignment to a Chicago ward. Ward and alderman assignments use the listing’s first observed date, with May 18 as the map transition date. The analysis excludes listings with coordinates that fail manual review, addresses that appear at inconsistent locations, and addressless listings concentrated at generic map points. It also excludes an address when replacing a listing’s coordinate with

that address’s most frequently recorded location changes its ward assignment or moves its measured boundary distance by more than 100 ft. Regressions also require square footage, bathrooms, and amenity controls. Bedrooms enter categorically, so studios remain in the sample.

B Alderman Fixed Effects and Index Validation

This appendix provides additional detail on how the aldermanic stringency index is constructed and validated.

B.1 Construction of the Stringency Index

Aldermanic stringency measures whether comparable high-discretion permits take systematically longer under a particular alderman. It uses the permit sample described in Section A.2 through December 2022. Permits must take at least one day to enter the processing-time comparison, although same-day permits still count toward the ward’s recent permit volume.

Stage 1: Adjusting Processing Times. I first adjust processing times for observable reasons that otherwise could make permits slower in some wards. The regression controls for the permit’s distance to downtown and Lake Michigan, nearby CTA stations, ward demographics, recent ward permit volume, calendar month, permit type, and review type. Appendix A describes how block group demographics are allocated across ward boundaries to construct the ward controls.

Stage 2: Alderman Differences. I average the adjusted processing times by ward and month, then compare these averages across the aldermen serving each ward-month. Formally, I regress the ward-month averages on alderman indicators $\alpha_{a(w,m)}$ and the same recent permit-volume control. Ward-months are weighted by their permit counts. Here $a(w, m)$ denotes the

alderman serving ward w in month m . The resulting fixed effect is the persistent alderman-specific difference in adjusted processing time.

Empirical Bayes Shrinkage. Some aldermen have fewer permits and therefore less precise estimates. I use empirical Bayes shrinkage to pull these noisier estimates toward the overall mean while leaving more precise estimates closer to their raw values: $\hat{\alpha}_a^{\text{EB}} = B_a \hat{\alpha}_a$, where $B_a = \hat{\omega}^2 / (\hat{\omega}^2 + \hat{\sigma}_a^2)$. Here $\hat{\omega}^2$ is the estimated variance across aldermen and $\hat{\sigma}_a^2$ is the sampling variance of alderman a 's estimate.

Index Standardization. The stringency index, S_a , is standardized to have mean zero and unit standard deviation across aldermen. Higher values indicate more stringent aldermen.

Table B.1 reports the adjustment regression used in the first stage. The dependent variable is permit-level log processing time. These controls remove predictable differences in timing across locations and permit types before I compare aldermen. Panel B of Figure B.1 shows the distribution of the stringency scores.

B.2 Validation of the Stringency Index

The stringency index should measure differences across aldermen rather than permanent differences across wards. Panel C of Figure B.1 compares predecessor and successor aldermen within the same ward. If fixed ward characteristics drove the scores, aldermen serving the same ward at different times would receive similar values. The relationship is weak, which weighs against that explanation.

Table B.1: Factors Associated with Permit Processing Times

	Log Processing Time (days) (1)
Median Household Income (\$10,000)	-0.015 (0.016)
Black Population Share	0.176 (0.278)
Hispanic Population Share	0.404 (0.272)
White Population Share	0.530 (0.352)
Homeownership Rate	0.261 (0.222)
Population (10,000)	-0.002 (0.020)
Distance to Downtown (km)	-0.017*** (0.006)
Distance to Lake (km)	-0.002 (0.005)
CTA Stations within 800 m	-0.009*** (0.003)
Permits in Previous Month	-0.002*** (0.0006)
Calendar Month Fixed Effects	✓
Permit Type Fixed Effects	✓
Review Type Fixed Effects	✓
N	120,505

Notes: Permit-level linear regression used to construct the stringency index, using permit records through December 2022. Covariates are median household income, racial shares, homeownership, population, distance to downtown, distance to the lake, nearby CTA stations, and the previous month's permit volume. Fixed effects are listed in the table. Regressions are unweighted. Standard errors are clustered by ward and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

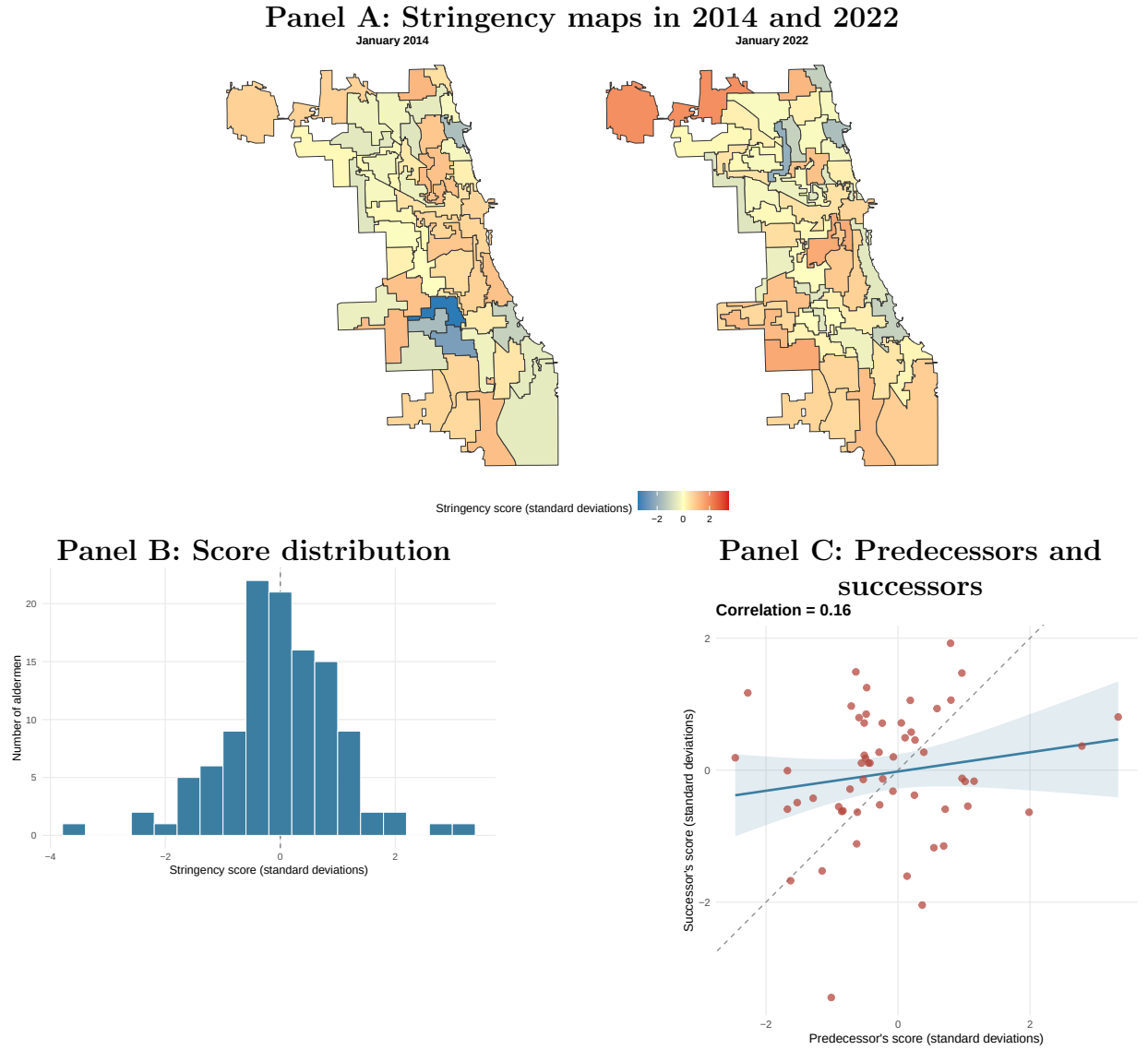


Figure B.1: Variation in Aldermanic Stringency

Notes: Stringency scores use permit processing times through December 2022 after adjusting for permit, location, ward, and calendar characteristics. Noisier alderman estimates are pulled toward the overall mean, and the resulting scores are standardized to mean zero and unit standard deviation. Panel A maps the scores of aldermen serving in January 2014 and January 2022 onto the corresponding ward boundaries; both maps use the same color scale, with warmer shades indicating more stringent aldermen. Panel B's dashed line marks the mean score of zero. Each point in Panel C pairs successive aldermen from the same ward. The dashed line indicates equal predecessor and successor scores; the solid line is a linear fit, with a shaded 95% confidence interval for the fitted mean.

C Additional Permit Evidence

This appendix reports additional permit evidence that complements the descriptive discussion in Section 2 and the event study in Section 4.1.

C.1 Balance Before Redistricting

Table C.1 compares blocks that changed wards in 2015 with nearby blocks that remained in the same ward. Before the new map took effect, the two groups were broadly similar in permit activity and neighborhood characteristics. The joint test across all listed characteristics has a p -value of 0.333.

Table C.1: Pre-Redistricting Balance Between Reassigned and Unchanged Blocks

Variable	Unchanged	Reassigned	Difference	SE	p -value	N
<i>Permit history</i>						
High-discretion permits, 2010–2014	2.668	2.649	0.086	0.081	0.291	2,340
Low-discretion permits, 2010–2014	4.278	4.377	0.146	0.412	0.725	2,340
No low-discretion permit, 2010–2014	0.114	0.127	0.008	0.023	0.713	2,340
New-construction permits, 2010–2014	0.381	0.349	0.013	0.044	0.766	2,340
No new-construction permit, 2010–2014	0.773	0.798	-0.008	0.013	0.525	2,340
<i>Neighborhood characteristics</i>						
Median household income	58,210	53,228	-1,352	1,354	0.324	2,324
Homeownership rate	0.404	0.407	0.000	0.011	0.998	2,340
Bachelor’s degree or higher share	0.441	0.399	-0.000	0.009	0.970	2,340
Black population share	0.301	0.272	0.003	0.006	0.667	2,340
Hispanic population share	0.226	0.276	-0.011	0.009	0.230	2,340
Median gross rent	1,068	1,022	-19	16	0.246	2,285
Median home value	309,400	283,787	-8,561	11,019	0.441	2,288
Average household size	2.550	2.694	-0.004	0.052	0.932	2,340
Median age	33.8	33.5	0.1	0.4	0.798	2,340
Population per square kilometer	8,350	8,187	-100	378	0.791	2,340

Notes: Reassigned blocks moved to a different ward under the 2015 map; unchanged blocks remained in their original ward. The sample requires the 2014 origin- and destination-ward incumbents to remain in office when the new map took effect. It also requires at least one high-discretion permit during 2010–2014. The sample contains 687 reassigned and 1,653 unchanged blocks within 500ft of a ward boundary in 44 ward pairs containing both groups. Permits are grouped by application year. Neighborhood characteristics are 2014 ACS five-year block-group estimates. Difference compares reassigned and unchanged blocks within the same ward pair; standard errors are clustered by ward pair. Means are unadjusted block-level means. The joint-test p -value for all listed covariates is 0.333 (complete-case $N = 2,249$).

C.2 Permit Event-Study Estimates

Table C.2 summarizes the average change in annual permit counts during the first six years under the new ward map. The combined model treats reassignment toward greater stringency and greater leniency as changes of equal size in opposite directions. A second model allows the two directions to have different effects. Low-discretion permits show a more muted response.

Table C.2: Permit Effects with Combined and Separate Reassignment Directions

	High-Discretion	Low-Discretion
Combined reassignment effect	-0.127** (0.049)	-0.017 (0.050)
<i>Directions estimated separately</i>		
Assigned to more stringent aldermen	-0.231** (0.096)	0.037 (0.074)
Assigned to more lenient aldermen	0.067 (0.059)	0.043 (0.065)
One-half difference between directions	-0.149*** (0.052)	-0.003 (0.048)
Equal-and-opposite test p -value	0.180	0.432
Block fixed effects	Yes	Yes
Ward-pair $\times$ year fixed effects	Yes	Yes
Positive pre-period permit activity required	Yes	Yes
N	36,830	35,541

Notes: Poisson regressions use census-block-year observations from 2010–2020 within 500 ft of a ward boundary. Stringency scores are estimated using permits filed from 2006 through 2014. The sample retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. It also requires at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified as assigned toward greater stringency, assigned toward greater leniency, or unchanged under the new map. The combined model imposes equal-and-opposite effects for the two reassignment directions, with unchanged blocks as the reference group; a negative coefficient means that greater stringency reduces permits. The separate model includes one indicator for each reassignment direction. Its half-difference row puts the two-direction estimates on the same scale as the combined estimate. The final test asks whether the separately estimated effects are equal in size and opposite in sign. Each model includes block fixed effects and ward-pair $\times$ year fixed effects. The estimator omits blocks with no permits of the relevant type in any sample year and ward-pair-year groups with no such permits. Standard errors, in parentheses, are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Allowing the Two Reassignment Directions to Differ

Panels B and C of Figure 2 allow moves toward greater stringency and greater leniency to have different effects. The two paths need not be mirror images, but their difference provides a comparable measure of how permit activity changes with stringency. Table C.2 shows that this comparison is nearly identical to the combined estimate.

Panel A of Figure C.1 shows the corresponding path for low-discretion permits. The response is more muted than for high-discretion permits.

C.4 Including Wards with Alderman Turnover

Panel B includes blocks from wards that elected new aldermen in 2015. The reassignment classification still uses the two aldermen serving in 2014. For blocks whose alderman subsequently changed, this measures the change implied by the map rather than the stringency the block necessarily experienced under the newly elected alderman. The annual number of high-discretion permits is 8.3% lower, a somewhat smaller estimate than in the main sample.

C.5 Permit Summary Statistics and Volume

Table C.3 compares the high- and low-discretion permit groups and reports their ward-month processing-time and permit-volume correlations. High-discretion permits are much slower than lower-discretion permits: the mean processing time is 57.3 days versus 26.2 days, and the median is 30 days versus 6 days. They also display more cross-alderman variation in processing times. Ward-months with longer high-discretion processing times have fewer high-discretion permits. Low-discretion permits show a weak positive relationship instead.

D Additional Density Boundary Checks

This appendix asks whether the density result could be an artifact of how I measure aldermanic stringency or where ward boundaries happen to lie. It also tests for jumps away

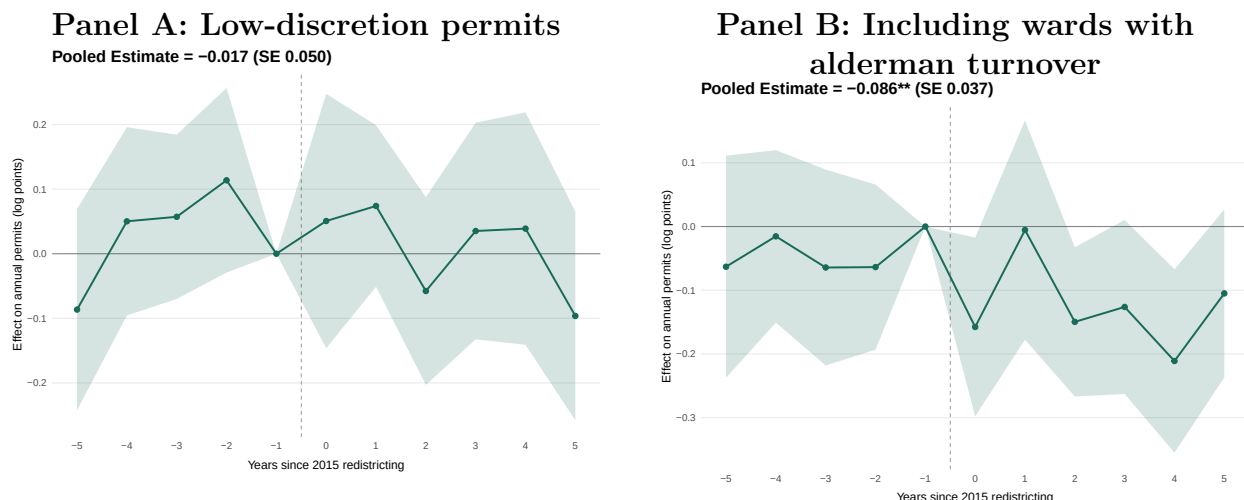


Figure C.1: Additional Permit Event-Study Checks

Notes: Panel A uses low-discretion permits excluding signs. Panel B uses high-discretion permits and includes wards with alderman turnover; assignment still uses the 2014 incumbents. Both panels combine the two reassignment directions using Equation (3); negative coefficients indicate fewer permits under assignment toward greater stringency. Points show effects in log points relative to 2014; subtitles report pooled 2015–2020 coefficients on the same scale. The pooled effects correspond to reductions of 1.7% and 8.3% in annual permits, respectively. Tests that the pre-2015 effects are jointly zero give $p = 0.075$ in Panel A and $p = 0.666$ in Panel B. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$.

Panel A: Processing-time summary statistics

Statistic	High-Discretion	Low-Discretion
Total permits in sample	121,405	124,124
Mean processing time (days)	57.3	26.2
Median processing time (days)	30.0	6.0
Alderman-level mean processing time (days)	60.2	26.6
Alderman-level IQR of mean processing time (days)	24.5	13.5
Number of unique wards	50	50
Number of unique aldermen	114	114

Panel B: Ward-month processing time and permit volume

Permit group	Correlation with ward-month permit volume
High-Discretion	-0.123
Low-Discretion	0.106
All	-0.021

Table C.3: Summary Statistics for High- and Low-Discretion Permits

Notes: The sample uses permits issued at least one day after application, with applications begun through December 2022. The permit groups are defined in Section 2.2. The alderman-level interquartile range is computed across alderman-specific mean processing times. Panel B reports Pearson correlations. Each observation represents one ward in one month.

from true boundaries and checks whether projects immediately next to a boundary drive the result. Unless otherwise noted, each estimate compares the two 100 ft bands immediately across the ward boundary using Equation (4). All checks use projects with both FAR and DUPAC available.

D.1 Stringency-Score Checks

The first check asks whether a project's own permits influence the score used to classify its boundary. For each project, I recalculate the scores for the two aldermen after removing permits matched to that project. The estimates are effectively unchanged. The second check drops boundaries where the two aldermen receive similar scores and the more-stringent side is therefore harder to distinguish. The estimates of FAR and units per acre remain negative in both construction samples when the required score difference is either 0.25 or 0.50 standard deviations, although the multifamily estimates become less precise as the sample shrinks.

Table D.1: Density Estimates Under Alternative Measures of Boundary Stringency

	All Construction		Multifamily	
	Log floor-area ratio	Log units per acre	Log floor-area ratio	Log units per acre
<i>Panel A: Removing permits linked to each project</i>				
Full-sample score	-.097** (.045)	-.147** (.072)	-.138* (.070)	-.191* (.108)
N	4,023	4,023	863	863
Ward pairs/segments	122/873	122/873	88/425	88/425
Score excluding the project's permits	-.097** (.045)	-.147** (.072)	-.138* (.070)	-.191* (.108)
N	4,023	4,023	863	863
Ward pairs/segments	122/873	122/873	88/425	88/425
<i>Panel B: Excluding boundaries with similar scores</i>				
Score difference at least 0.25 SD	-.118** (.059)	-.171** (.078)	-.066 (.084)	-.070 (.109)
N	2,964	2,964	710	710
Ward pairs/segments	118/771	118/771	80/357	80/357
Score difference at least 0.50 SD	-.122* (.064)	-.172** (.085)	-.153 (.105)	-.141 (.124)
N	2,313	2,313	530	530
Ward pairs/segments	100/582	100/582	64/273	64/273

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The second row removes permits matched to each construction project before recalculating the alderman scores used for that project. The score-difference rows retain boundaries where the absolute difference between the two alderman scores is at least the stated number of standard deviations. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 Placebo Cutoffs

Figure D.1 moves the cutoff 1,000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. None of the eight placebo estimates is statistically distinguishable from zero at the 10% level.

D.3 Continuity in Location Characteristics

Table D.2 tests whether fixed features of a project’s location change at the ward boundary. None of the differences in distance to downtown, the nearest school, the nearest park, or Lake Michigan is statistically distinguishable from zero in either sample.

Table D.2: Continuity in Location Characteristics at Ward Boundaries

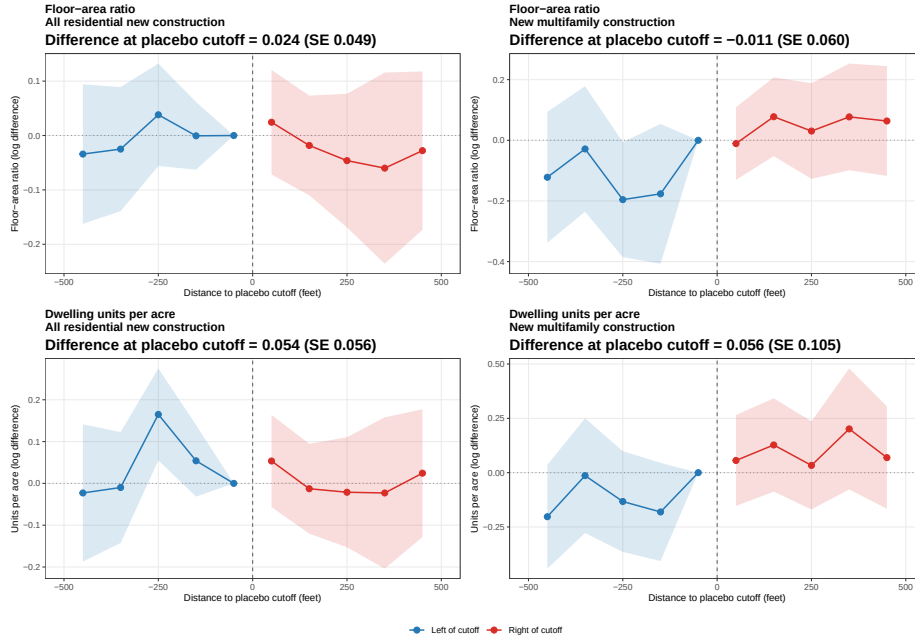
	All Construction	Multifamily
Distance to downtown (miles)	0.001 (0.009)	-0.009 (0.012)
Distance to nearest school (miles)	0.007 (0.005)	-0.002 (0.009)
Distance to nearest park (miles)	-0.006 (0.006)	-0.011 (0.010)
Distance to Lake Michigan (miles)	0.007 (0.007)	0.002 (0.011)
Observations	4,023	863

Notes: Each row uses the listed location characteristic as the dependent variable. The reported coefficient compares the 0–100 ft band on the more-stringent side with the –100 to 0 ft band on the less-stringent side. Regressions include 100 ft distance-band indicators, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.4 Boundary Geometry

Some ward boundaries follow expressways, waterways, arterial roads, or other physical features. Table D.3 therefore repeats the analysis after restricting the sample based on the geometry of the assigned boundary segment. Removing segments with substantial expressway or water overlap leaves the estimates almost unchanged. A broader physical-feature and

Panel A: 1,000 ft inside the less-stringent side



Panel B: 1,000 ft inside the more-stringent side

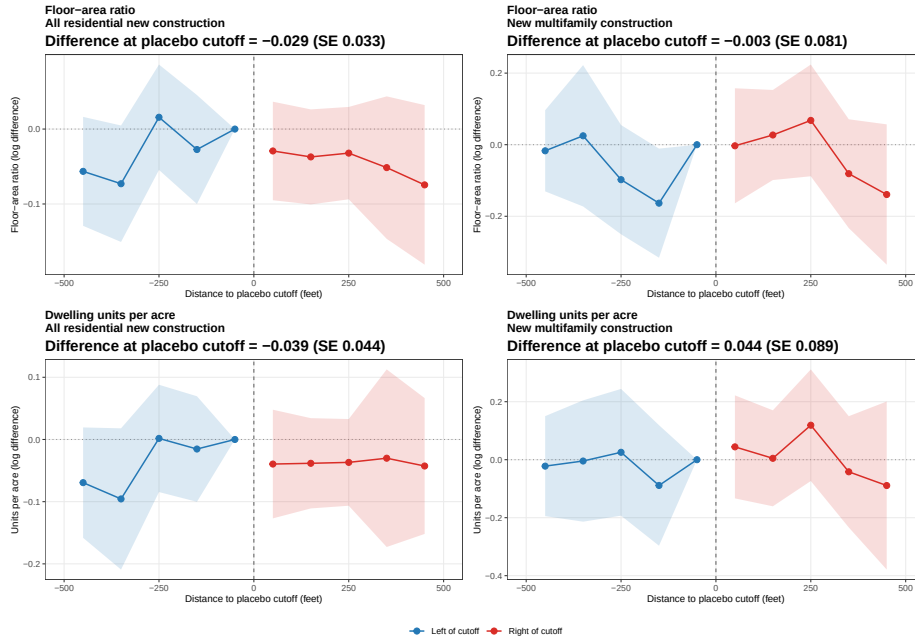


Figure D.1: Density Estimates at Placebo Cutoffs

Notes: Each artificial cutoff lies 1,000 ft inside the indicated side of the true ward boundary. The model uses a 500 ft window around the artificial cutoff and the same 100 ft distance bands, controls, and fixed effects as Equation (4). The 100 ft band just below the artificial cutoff is the reference group; subtitles report the difference for the 100 ft band just above it. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

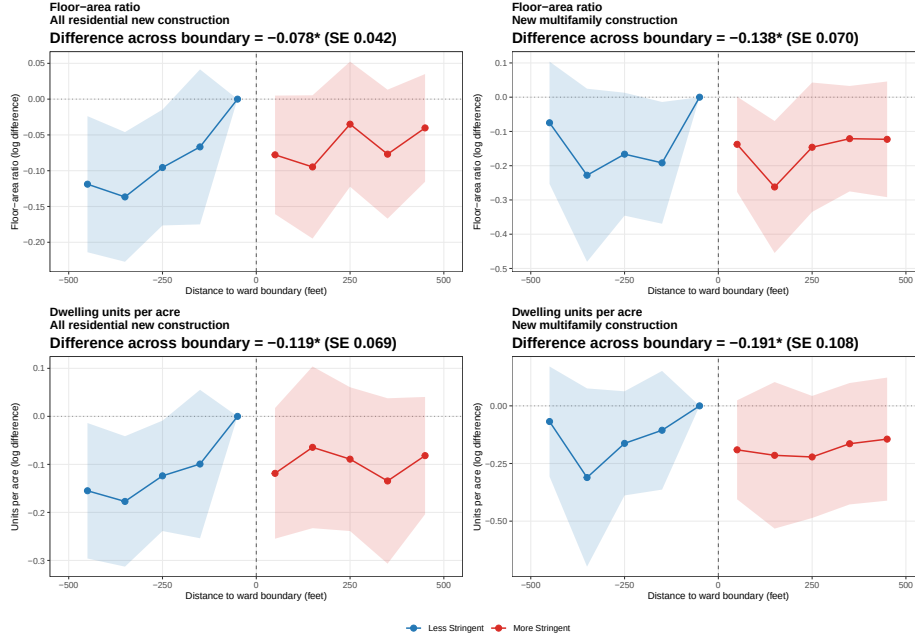
arterial-road restriction attenuates the multifamily estimates, while the all-construction estimates remain negative. Restricting the sample to locally straight boundaries also produces negative estimates for FAR and units per acre in both construction samples. These estimates are generally smaller and less precise in the reduced sample, but they do not suggest that boundary curvature alone creates the main pattern.

Table D.3: Density Estimates Along Different Types of Ward Boundaries

	All Construction		Multifamily	
	Log floor-area ratio	Log units per acre	Log floor-area ratio	Log units per acre
Main sample	-0.097** (0.045)	-0.147** (0.072)	-0.138* (0.070)	-0.191* (0.108)
Observations	4,023	4,023	863	863
Limited expressway or water overlap	-0.096** (0.045)	-0.146** (0.072)	-0.139* (0.070)	-0.193* (0.109)
Observations	3,769	3,769	840	840
Limited physical-feature or arterial overlap	-0.082* (0.045)	-0.128* (0.068)	-0.135* (0.078)	-0.152 (0.096)
Observations	3,181	3,181	690	690
Straight boundary segment	-0.081* (0.046)	-0.125* (0.065)	-0.109 (0.081)	-0.161 (0.121)
Observations	3,140	3,140	625	625

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The expressway-or-water restriction removes segments for which either feature overlaps at least half of the segment. The broader restriction removes segments with at least 50% park, water, or cemetery overlap, 40% expressway overlap, or 75% arterial-road overlap. A boundary is classified as locally straight when a 328 ft straight line centered on the project's nearest boundary point stays within 49 ft of the actual boundary at both ends. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Excluding projects within 25 ft



Panel B: Excluding projects within 50 ft

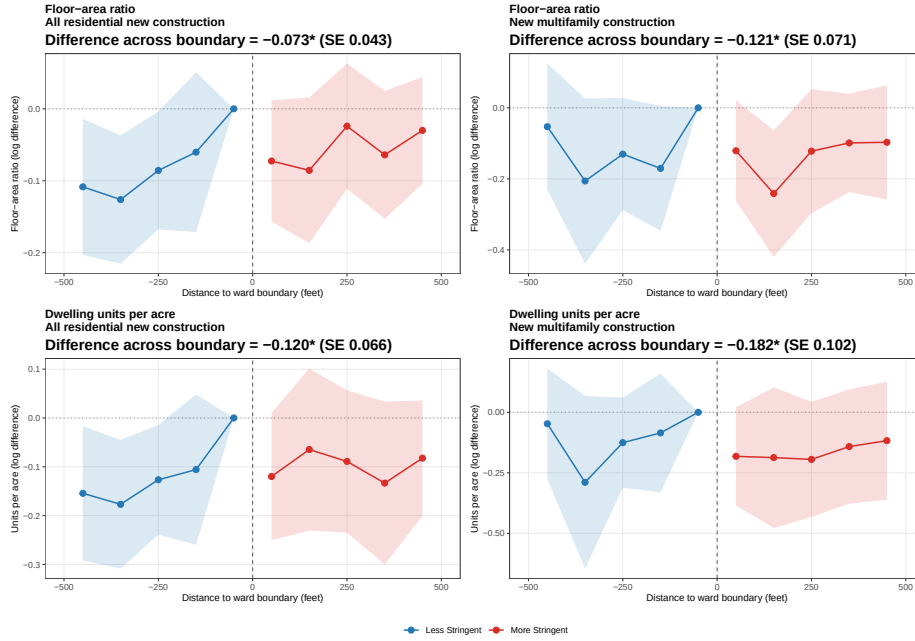


Figure D.2: Density Estimates Excluding Projects Nearest the Boundary

Notes: This check asks whether projects immediately adjacent to a boundary drive the main result. Each panel excludes projects within the indicated distance and otherwise uses the 500 ft sample and Equation (4). After each exclusion, the remaining part of the -100 to 0 ft band on the less-stringent side is the reference group. In each panel, the estimated difference across the boundary remains negative for FAR and units per acre in both construction samples. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Price Boundary Checks

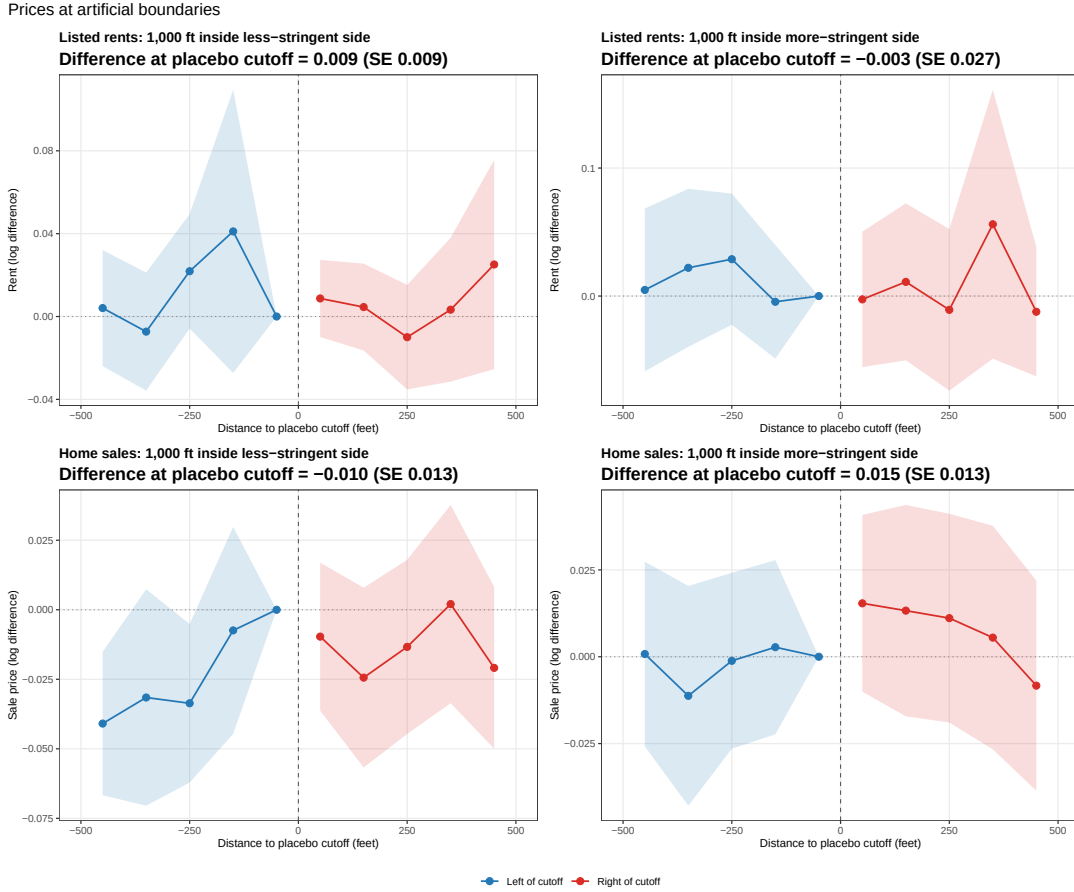
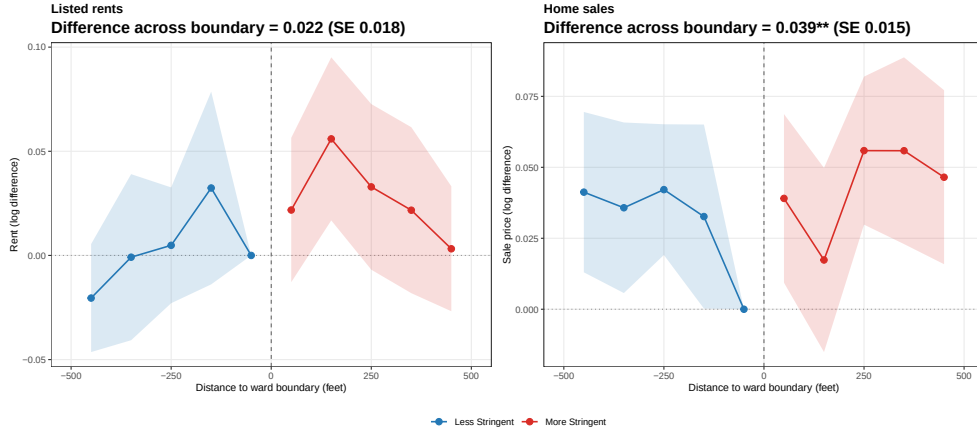


Figure E.1: Placebo Cutoffs for Listed Rents and Home Sale Prices

Notes: Neither rents nor sale prices change significantly across an artificial cutoff. The left column places an artificial cutoff 1,000 ft inside the less-stringent ward; the right column places it 1,000 ft inside the more-stringent ward. The top row reports listed rents and the bottom row reports home sale prices. Each model uses a 500 ft window around the artificial cutoff and otherwise follows Equation (5). Each point is the estimated log difference between a 100 ft distance band and the band immediately below the artificial cutoff. Subtitles compare the bands immediately above and below the cutoff. Shaded areas are 95% confidence intervals. Standard errors are clustered by ward pair.

Panel A: Locally straight boundaries



Panel B: Excluding observations nearest the boundary

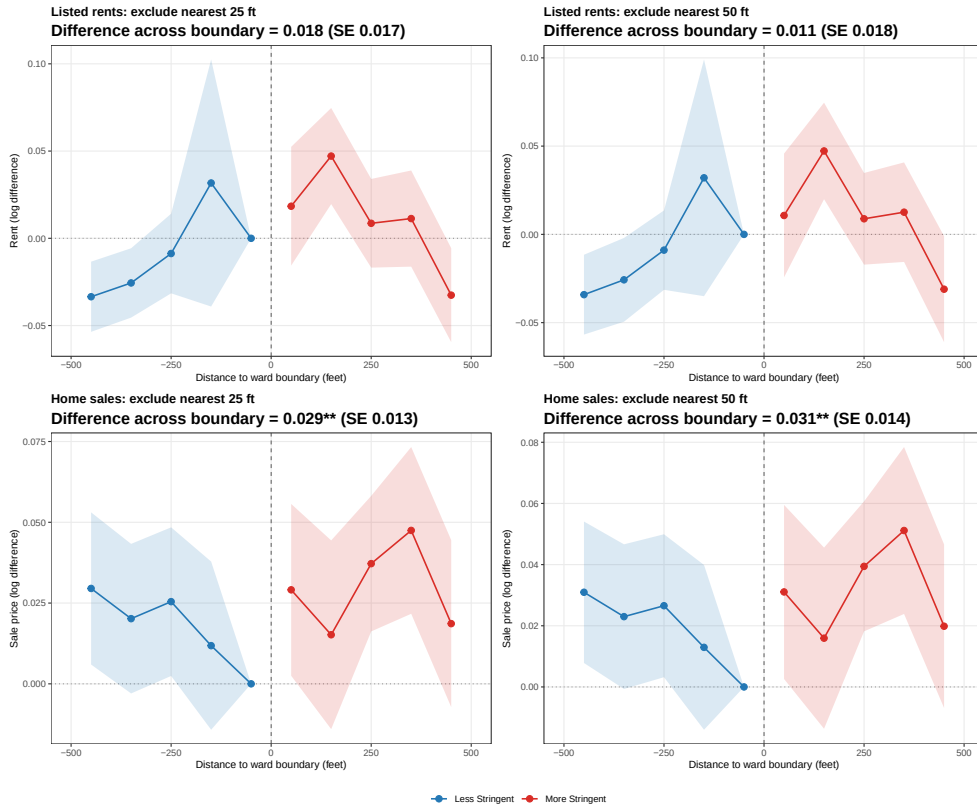


Figure E.2: Additional Price-Boundary Robustness Checks

Notes: Panel A retains locally straight boundaries: a 328 ft line centered on the nearest boundary point stays within 49 ft of the boundary at both ends. Panel B excludes observations within 25 ft (left) or 50 ft (right); the top row reports listed rents and the bottom row home sale prices. All models otherwise follow Equation (5). Points report log differences from the nearest 100 ft band on the less-stringent side, normalized to zero; subtitles compare the bands across the boundary. Panel B uses the remaining observations in each band after exclusion. Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.